# PayResolve AI: Executive Research & Roadmap

## Autonomous Multi-Agent Dispute Resolution & Settlement Swarm for the PayPal AI Hackathon

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| **Target Competition:** PayPal AI Hackathon 2026 | **Prize Target:** Grand Prize ($12k) + Agentic Commerce ($5k) |
| **Core Integration:** PayPal REST API & Sandbox + LLM Swarm | **Status:** Research Verified & Scaffold Ready |

## 1. Executive Summary & Market Problem Space

**The $100 Billion Friction Point:** In modern e-commerce, customer disputes, claims, and chargebacks represent one of the highest friction operating costs for online merchants and payment platforms. When a buyer encounters an issue (package delayed, item damaged, or unrecognized charge), current customer support systems trap them in multi-day ticket queues. If unaddressed, buyers escalate issues into formal PayPal disputes or credit card chargebacks.

**The Merchant Dilemma:** Every escalated dispute risks a $20+ administrative fee, lost inventory, payment holdbacks, and gateway reputation penalties. Merchants manually cross-reference shipping tracking, check store policies, and log into their payment portal to issue refunds—a slow, expensive, and error-prone process.

**The Opportunity:** 80% of customer disputes occur during the initial *Inquiry Phase* before formal claims are filed. By deploying an autonomous, multi-agent AI system directly integrated with the PayPal Developer Platform, merchants can triage, investigate, risk-score, and settle valid disputes in seconds, slashing chargeback rates by over 70%.

## 2. Deep Dive: PayPal Dispute Lifecycle & API Integration

Official research into the PayPal Developer Platform confirms a highly structured dispute lifecycle that PayResolve AI automates:

| Lifecycle Phase | PayPal Technical Mechanism | PayResolve AI Autonomous Action |
|---|---|---|
| **1. Pre-Claim / Inquiry** | Direct buyer-seller negotiation window (up to 20 days) via v1/customer/disputes. | Customer Intake Agent engages buyer immediately, gathers photos/receipts, preventing formal escalation. |
| **2. Verification & Audit** | Query transaction details via /v2/payments/captures/{id}. | Evidence Agent queries carrier APIs while Fraud Scorer audits buyer history and computes risk. |
| **3. Instant Settlement** | Trigger refund via /v2/payments/captures/{id}/refund or accept claim. | PayPal Settlement Agent issues automated full/partial refund in sandbox, logging complete audit trace. |

## 3. Why a Multi-Agent Swarm? (The Architectural Edge)

Single-prompt LLMs fail in financial and legal environments because they are vulnerable to prompt injection, emotional bias, and hallucinations. PayResolve AI uses a **Hub-and-Spoke Multi-Agent Topology** where each agent has strict boundaries:

- **Customer Intake Agent:** Empathetically communicates with the customer, collects issue category, photos, and tracking numbers. Has zero authority to touch money.
- **Evidence & Tracking Agent:** Simulates/calls carrier APIs to verify actual delivery status, carrier GPS drop coordinates, and delivery timestamps.
- **Fraud & Risk Scoring Agent:** Computes an objective risk score (0.0 to 1.0) using order velocity, transaction size, and historical dispute anomalies.
- **Policy & Terms Auditor:** Validates merchant warranties, return policies (e.g., 30-day window, perishable goods exemptions), and allowable concessions.
- **Supervisor Arbiter:** Synthesizes inputs from all workers. If Risk < 0.20 and Evidence is verified, signs off on auto-settlement; otherwise escalates to merchant.
- **PayPal Settlement Agent:** Executes the actual cryptographic / API call to PayPal Sandbox to trigger captured payment refunds or dispute settlements.

## 4. Step-by-Step Implementation Roadmap

To ensure a seamless, high-scoring hackathon submission within our 11-day timeframe, we execute across 5 structured phases:

| Phase | Deliverable Scope | Status / Verification |
|---|---|---|
| **Phase 0: Setup** | Devpost draft, Git repo, MIT license, PayPal Sandbox API credentials & virtual environment. | **COMPLETED & VERIFIED** |
| **Phase 1: Backend** | FastAPI endpoints, PayPal REST client (Captures, Refunds, Disputes), mock carrier tracking. | Next to execute |
| **Phase 2: Swarm Core** | Gemini 2.5 Flash agent tools, structured schema outputs, risk scoring algorithm, arbiter synthesis. | Planned (Ready to build) |
| **Phase 3: Visualizer UI** | React + Vite + Tailwind CSS: Split screen with buyer dispute chat and live multi-agent DAG reasoning view. | Planned |
| **Phase 4: QA & Polish** | Automated test scenarios (clean claim, partial refund, fraudulent claim, human escalation). | Planned |
| **Phase 5: Submission** | 3-minute demo video (YouTube), Devpost final sync, GitHub README screenshots. | Target: Before Nov 12 |

## 5. Hackathon Winning Strategy (Evaluation Mapping)

Each component directly maps to the hackathon's 5 official 20% judging criteria:

1. **Technical Implementation (20%):** Direct integration with live PayPal Sandbox REST APIs (v2 captures, v2 refunds, v1 disputes) paired with multi-agent LLM tool execution.

2. **Design & UX (20%):** A high-polish dual dashboard: customer sees a fluid conversational helper, while the merchant sees live agent collaboration and financial analytics.
3. **Potential Impact (20%):** Targets a massive e-commerce problem—reducing dispute handling from 14 days down to 60 seconds and saving merchant chargeback fees.
4. **Innovation & Agentic Commerce (20% + $5k Prize):** Pioneering true autonomous financial agency with safe, human-in-the-loop guardrails.
5. **Presentation (20%):** A crisp, highly visual 3-minute video showing a simulated buyer claim being autonomously triaged and refunded live on PayPal sandbox.